Lab 3 Architecture Justification

Self-Service Coffee Kiosk | Component Modelling & Architectural Pattern Selection

We chose a layered architecture for the kiosk. It separates the touch interface, order workflow, payment integration, and menu/pricing data while keeping the system simple to deploy as one kiosk application.

Architecture options

Style	Fit for this kiosk	Trade-off
Layered (selected)	Clear UI, business, payment, and data boundaries for one kiosk.	Low deployment and maintenance overhead. Changes can still couple layers.
Microservices	Independent services can scale and deploy separately.	Network, monitoring, and service-operation costs are excessive for one kiosk.
Client-server	A kiosk client can request central menu and payment services.	Useful across many kiosks, but depends on connectivity and does not define internal separation as clearly.

Why layered suits the scenario

Scenario fit. The kiosk has a focused workflow: select Espresso, Americano, or Latte; choose Small or Large; review the total; pay by card; and print a receipt. The Order Manager coordinates that flow while the menu repository supplies prices.

Security. The Payment Service sends tokenized authorization requests to the card gateway over HTTPS/TLS. The Order Manager receives a result or token and does not handle or log raw card numbers. Only payment-related components need the gateway integration.

Performance. The menu and prices are read locally, avoiding a network round trip for each screen or price lookup. Card authorization is requested only at checkout; the kiosk waits for its result before confirming the order and printing the receipt.

Model: Touchscreen UI → Order Manager → Menu & Pricing Repository / Payment Service / Receipt Printer Adapter; external interfaces connect to the card gateway and thermal printer. See component_diagram.png.